

Waste Whisper — Zero se Hundred tak Building Guide (Hinglish)

Ye guide tumhare 4-5 mahine ke timeline ke hisaab se hai. Order follow karo, phase jump mat karo — hardware project mein sabse zyada time waste tab hota hai jab log sab kuch ek saath shuru kar dete hain.

PHASE 0: Planning aur Shopping (Week 1)

Sabse pehle ye karo, koi part order mat karo abhi:

1. Poore team ke saath baith ke plan discuss karo — kisko kya karna hai (AI/ML, Embedded, Mechanical, Backend) wo pehle hi decide kar lo, jo pichle plan mein bataya tha.
2. Ek **shared Google Sheet/Drive folder** bana lo — sabka kaam, progress, aur parts list ek jagah track ho.
3. Poori parts list nikaal lo (jo humne pehle discuss kiya) aur local electronics market ya online (Robu, Robocraze, Amazon) se **ek saath order karo** — alag-alag baar order karne se shipping cost zyada lagta hai.
4. College coordinator/lab se pooch lo — agar ESP32, sensors, ya conveyor parts lab mein already available hain to unko use karo, paisa bacha.

Is week ka goal: Sab parts order ho chuke hon, aur sabko pata ho apna role kya hai.

PHASE 1: Mechanical Base Banao (Week 2-3)

Ye sabse pehle banega kyunki baaki sab isi ke upar depend karta hai.

1. Conveyor ka frame banao — plywood ya acrylic se. Agar budget tight hai to plywood + PVC pipe rollers use karo.
2. Belt lagao — ya to ready-made conveyor belt kharido, ya purane treadmill ka rubber belt use karo (sasta option).
3. DC motor ko frame pe mount karo, aur motor driver (L298N) se connect karo.
4. **Sabse important:** Belt ko manually test karo — bina kisi AI/sensor ke, sirf motor on/off karke dekho belt smoothly chal rahi hai ya nahi. Agar belt hilti hai, roller align nahi hai, to yahiin fix karo — baad mein fix karna mushkil ho jayega.

Is phase ka goal: Belt reliably chale, bina jaam hue, motor driver se control ho sake.

PHASE 2: Camera-Only Classifier Banao (Week 3-5)

Ye parallel mein chal sakta hai Phase 1 ke saath (AI/ML member ye start kar sakta hai jab mechanical member frame bana raha ho).

1. Pehle ek **dataset collect karo** — Paper, Plastic, Metal ki 100-150 photos har category ki. Apne phone se alag-alag angle, lighting mein photos khincho. (Ya online dataset jaise TrashNet bhi use kar sakte ho, phir apne khud ke photos add karo.)

2. **Google Colab** use karo training ke liye — free GPU milta hai, laptop ka load nahi padega.
3. Ek **lightweight model** train karo — MobileNetV2 ya ResNet18 pe transfer learning karo (scratch se training mat karo, time waste hoga aur accuracy bhi kam aayegi).
4. Model ko laptop pe load karke webcam se live test karo — Python + OpenCV se camera feed lo, model se predict karwao.
5. **Confidence score bhi nikaalo** — sirf label nahi, ye bhi dekho model kitna "sure" hai (softmax probability). Ye baad mein confidence-gating ke liye zaroori hai.

Is phase ka goal: Laptop pe webcam se live object dikhao aur model sahi class + confidence score de raha ho, 80%+ accuracy ke aas paas.

PHASE 3: ESP32 aur Servo Gates (Week 4-6)

Embedded member ye start kar sakta hai jaise hi mechanical base ready ho (Phase 1 ke baad ya parallel mein).

1. ESP32 ko Arduino IDE/PlatformIO se setup karo, basic "blink" test karke pehle confirm karo board sahi se kaam kar raha hai.
2. Servo motors ko ESP32 se connect karo — pehle simple code se test karo ki servo sahi angle pe move ho raha hai.
3. IR/photoelectric sensor lagao belt ke start mein — jab object aaye to ESP32 ko signal jaye.
4. Ek basic logic likho: "jab IR sensor trigger ho, to ek fixed time baad servo gate open ho jaye" — abhi ye manual/fixed hai, AI se connect nahi hua.
5. Poora mechanical + electronics ka **dry run** karo — ek object belt pe daalo, IR trigger ho, servo sahi time pe move kare.

Is phase ka goal: Object belt se pass ho, sensor detect kare, servo sahi time pe gate open kare — bina AI ke, sirf timing-based.

PHASE 4: Physical Sensors Wire aur Calibrate Karo (Week 6-8)

Ye phase sabse zyada time lega, patience rakhna.

1. Inductive sensor lagao — metal ke paas laake test karo, reading dekho.
2. Capacitive sensor lagao — alag-alag material (plastic, paper) ke paas test karo, readings note karo.
3. Load cell + HX711 lagao — known weight (jaise 100g ka object) rakhke calibrate karo taaki accurate weight aaye.
4. Har sensor ki readings ko ek simple table mein likho — "Metal ke liye inductive sensor kya value deta hai", "Plastic ke liye capacitive sensor kya deta hai" — ye data baad mein fusion logic banane mein kaam aayega.
5. Ultrasonic sensors bin ke andar lagao — bin khaali hai ya bhara hai, ye detect karne ke liye.

Is phase ka goal: Har sensor consistent aur reliable readings de raha ho, aur tumhare paas ek reference table ho ki kaun sa material kaisi reading deta hai.

PHASE 5: Confidence Gating + Sensor Fusion Logic (Week 8-10)

Ye project ka "dimaag" hai — yahan AI aur sensors dono milke kaam karte hain.

1. Ek threshold decide karo — jaise agar model confidence 85%+ hai to seedha accept karo (auto-sort), 50-85% hai to sensors se extra evidence lo, 50% se kam hai to seedha "Uncertain/Manual" bin mein bhejo.
2. Medium-confidence cases ke liye — sensor readings ko camera prediction ke saath combine karo. Simple rule-based logic se start karo (jaise "agar inductive sensor high reading de raha hai to Metal maano, chahe camera confidence kam ho").
3. Isko Python code mein likho jo camera prediction + sensor data dono lekar final decision de.
4. Final decision ko ESP32 ko bhejo (Serial connection se abhi ke liye) taaki sahi servo gate trigger ho.

Is phase ka goal: System khud decide kare — kab sirf camera pe trust kare, kab sensors bhi check kare, kab reject karke manual bin mein bheje.

PHASE 6: IoT Dashboard aur Logging (Week 9-11)

Backend member ye phase lead karega, but ye Phase 5 ke saath parallel bhi chal sakta hai.

1. SQLite database banao — har object ka prediction, confidence, sensor readings, final class, aur time store karo.
2. FastAPI/Flask se ek simple backend banao jo data ko database mein save kare.
3. Dashboard banao (Streamlit sabse fast hoga banane mein, ya React agar zyada polish chahiye) — live throughput, class-wise count, confidence graph, bin fill level dikhao.
4. Ye backend ko ESP32 aur AI system dono se connect karo taaki real-time data aaye.

Is phase ka goal: Ek live dashboard chal raha ho jo dikhaye system ne kitne objects sort kiye, kaunsi class mein, aur bins kitne bhare hain.

PHASE 7: Poora Integration (Week 10-12)

Ab sab pieces ko jodna hai — ye sabse critical phase hai.

1. Poore team ke saath ek din baitho aur end-to-end test karo — object belt pe daalo, camera classify kare, confidence check ho, zaroorat pade to sensors trigger hon, servo sahi gate kholta hai, aur dashboard pe data live update hota hai.
2. **20-30 baar consecutively test karo** — dekho kitni baar system sahi kaam karta hai. Jo bhi fail ho, note karo aur fix karo.

3. Edge cases test karo — jaise composite material (chip packet), bahut chhota object, do objects ek saath — dekho system kaise react karta hai.

Is phase ka goal: System bina crash hue, bina manual intervention ke, khud se lagataar objects sort kar sake.

PHASE 8: Human-in-the-Loop + Experiments (Week 12-13)

Ye tumhara "research angle" hai jo project ko normal se alag banata hai.

1. Manual bin mein jo objects gaye hain, unko label karo (Paper/Plastic/Metal manually mark karo).
2. Ek chhota script banao jo ye labeled data future retraining ke liye save kare.
3. **Comparison experiment karo:** same test objects pe teen tarike se test karo — (a) sirf camera, (b) sirf sensors, (c) dono combine (fusion). Accuracy, wrong-sort rate, aur auto-accept rate compare karo.
4. Ye numbers ek chhoti table/graph mein likho — final report ke liye ye sabse important section hoga.

Is phase ka goal: Data-backed proof ho ki multimodal fusion camera-only se better hai.

PHASE 9: Polish aur Demo Prep (Week 13-15)

1. Poora system ek baar aur reliability check karo — demo se pehle sab kuch stable hona chahiye.
2. Ek **backup plan** rakho — agar koi sensor demo ke din fail ho jaye to kya karoge (jaise manual override button).
3. Presentation banao — architecture diagram, results, aur live demo ka flow decide karo.
4. Poore team ke saath demo ka **rehearsal** karo — kaun kya bolega, kaun kya operate karega.
5. Final report likho — sab phases, results, aur limitations clearly document karo.

Is phase ka goal: Confident demo, backup plan ready, aur report submission-ready.

Quick Order Cheat-Sheet (Yaad rakhne ke liye)

1. Planning + Shopping
2. Mechanical frame + belt (test manually)
3. Camera classifier (parallel)
4. ESP32 + servo (timing-based, bina AI ke)
5. Physical sensors wire + calibrate
6. Confidence gating + fusion logic (AI + sensors jode)

7. IoT dashboard

8. Full integration + reliability testing

9. Human-in-the-loop + comparison experiments

10. Polish + demo rehearsal

Golden rule: Har phase ke end mein ek chhota integration test zaroor karo, sirf apna hissa bana ke mat baith jao. Jo team last week mein sab jodne ki koshish karti hai, wahi fail hoti hai.